



Alien Guitar Secrets

The Essence of Time and Rhythm

An Introduction to Counting time in Music
and the development of Rhythm

by Rob Lobasso





Alien Guitar Secrets

TABLE OF CONTENTS

Introduction.....	2
Tempo.....	3
Note and Rest Values.....	4
Note Values	6
Whole notes	6
Half Notes	6
Quarter Notes.....	7
Eighth Notes	7
Sixteenth Notes.....	7
Rest Values.....	8
Whole Note Rests.....	8
Half Note Rests.....	8
Quarter Note Rests	9
Eighth Note Rests	9
Sixteenth Note Rests	9
Triplets	10
How To Count Beats	10
Time Signatures.....	12
Time Signatures In Action.....	14
Dream Theater - The Dance Of Eternity Excerpt.....	15
Essential Rhythmic Patterns.....	15
Rhythmic Pattern Practice.....	16
The TripletOr Swing Feel.....	Error! Bookmark not defined.



Alien Guitar Secrets

INTRODUCTION

Have you ever struggled with the Time and Rhythmic aspects of music?

For some, understanding scales, note pitch, chords etc. is their grind. But for others, it is Time and Rhythm, that stumps them.

This chapter has been written to hopefully give you a more concise, if not thorough understanding of this foundational principle, so that you may become confident in your understanding of Rhythm and Time. This, I believe is vital to the continued development of a guitarist.

In this document, we will discuss numerous aspects of Time such as:

- Tempo
- Note and Rest Values
- Time Signatures in Action
- Simple Time & Compound Time
- Straight v Swing
- Common Rhythmic Patterns
- Locking In! the Importance of Groove

So, buckle up as we embark on a rollercoaster ride as we explore the wild twists & turns in Unveiling 'The Essence of Time and Rhythm.



Alien Guitar Secrets

TEMPO

Tempo, is literally Latin for Time. In music, Tempo indicates the Pulse and therefore the speed of the music.

Tempo is measured in beats per minute (BPM), so you could think of it as a ticking clock. A marker which helps everything relating to a piece of music synchronise together and stay in time. But unlike a typical clock which is fixed at 60 seconds (ticks) per minute, the tempo in music is adjustable to any speed you wish within a minute, as an example you could have it slow at 60 beats per minute or extremely fast at 180 BPM.

Each pulse is generally measured in quarter note values (we will talk about this next)

You can hear and see this demonstrated when watching a live band. Someone within the band (typically the drummer) will count in the rest of the band with something along the lines stick clicks of 1-2-3-4. This is done to indicate to the rest of the band the tempo (speed) at which the song will be played at.

Staying in time for a lot of people can be quite a challenge, partially because as emotional beings, we tend to naturally speed up as we get into things or become excited by them, and music is no different. And that sure doesn't help the band stay in time together, nor does it help your playing when you're trying to record your parts. So, NOT practicing to a beat at a consistent tempo is the other reason people find it challenging, and therefore is a must to learn.

If you're a little lazy with your time, just deciding to make an exercise whereby you pick and fret notes or chords to a beat marker like 80 BPM can really help to get you confident over the beat. Even something as simple as this, is a good first step, and you will see improvement almost immediately.



Alien Guitar Secrets

The more you practice playing at tempo, the easier it becomes to find space within each pulse, which in turn leads to more creativity with the beat.

NOTE AND REST VALUES

If tempo is the constant speed of the pulse by which music is measured, then how do we vary the speed at which music is played, all the while relating to the Tempo of the music?

Well, we use a system of unique Notation and Rest symbols and assign different values to each of them. This helps us indicate the various speed changes within a piece of music. We call these Note and Rest Values, Subdivisions.

These subdivisions work by dividing the values of notes and rests into smaller values. It is very mathematical, using fractions as a means measurement and therefore, making it logical.

In visual terms it would be like getting a whole Pizza, and dividing it in half, then half again, and so on.

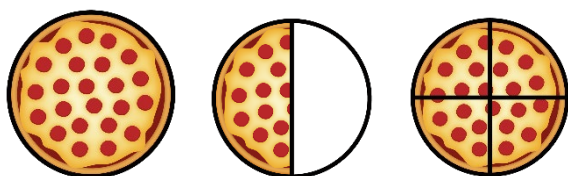


Figure 1. The dividing of a Pizza into slices as representation of time in music.

My aim is here is to make it as clear as I can, because I can remember being one of those people stumped by the following.

The Table over the page indicates the more common Note and Rest symbols used, their names and their Values.



Alien Guitar Secrets

It is important that you study, practice and memorise these as soon as you can. The quicker you can count and hear the differences between the said note and rest values, the sooner your fears about finding and moving to a beat will disappear.

Study the symbols for both Notes and Rests. Commit their Beat Values to memory.

Note Value	Note Symbol	Rest Symbol	Beat Value
Whole Note			4 Beats
Half Note			2 Beats
Quarter Note			1 Beat
Eighth Note			1/2 Beat
Sixteenth Note			1/4 Beat

Figure 2. Note and Rest Value Diagram

So, let's now break down each Subdivision, so we can get a better understanding of how to count these Time structures.



Alien Guitar Secrets

NOTE VALUES

A notes' value is simply the duration of beats that note is sustained for. The humble quarter note (crotchet) is the most used note value. The value given to the quarter note is one beat. Four of these quarter notes will be the equivalent of a Whole Note, two Quarter Notes, the equivalent of one Half Note etc.

WHOLE NOTES

Whole notes are sustained for exactly four quarter note beats as indicated below:

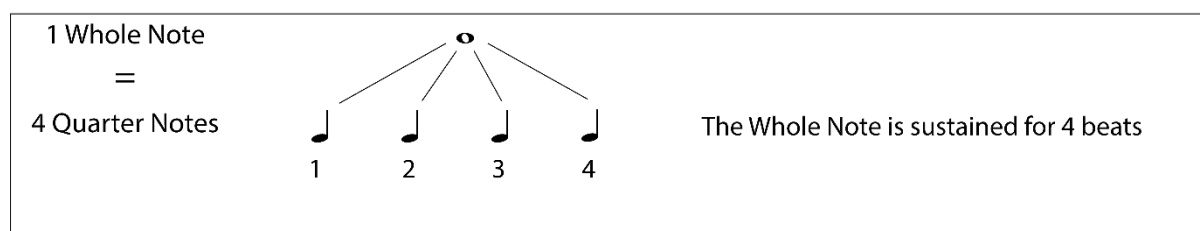


Figure 3. Whole Note Value

HALF NOTES

Half notes are sustained for exactly two quarter note beats as indicated below:

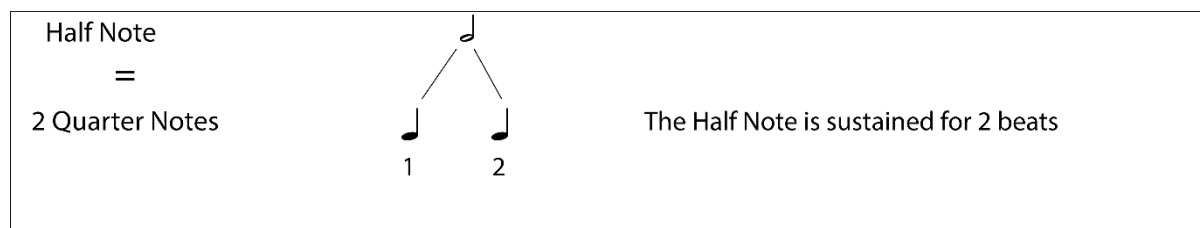


Figure 4. Half Note Value



Alien Guitar Secrets

QUARTER NOTES

Quarter notes are sustained for exactly 1 beat as indicated below:

Quarter Note =  and is sustained for 1 beat and counted as follows:    

1 2 3 4

Figure 5. Quarter Note Value

EIGHTH NOTES

Eighth notes are sustained for exactly $\frac{1}{2}$ (one half) of one beat and are indicated as follows:

Quarter Note =  The Eighth Note is sustained for 1/2 beat and can be counted as follows:    

2 Eighth Notes   1 a   2 a   3 a   4 a

Figure 6. Eighth Note Value

SIXTEENTH NOTES

Sixteenth notes are sustained for exactly $\frac{1}{4}$ (one quarter) of one beat value of and are indicated as follows:

Quarter Note =  The Sixteenth Note is sustained for 1/4 beat and can be counted as follows:    

4 Sixteenth Notes     1 e & a     2 e & a     3 e & a     4 e & a

Figure 7. Sixteenth Note Value



Alien Guitar Secrets

REST VALUES

As the name implies, a rest is a block of musical time in which a musician does not play a sound. Therefore, Rests are the complete opposite to Notes. They are however, counted in the same manner, even using beats as a unit measurement of the length of time one does not play for.

WHOLE NOTE RESTS

Whole Note Rests are a block of time you DONOT play for, and which last for exactly Four Quarter Note Beats.

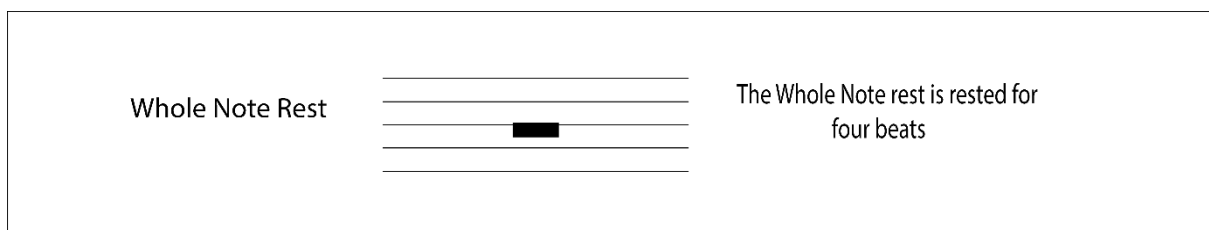


Figure 8. Whole Note Rest Value

HALF NOTE RESTS

Half Note Rests are a block of time you DO NOT play for, and which last for exactly Two Quarter Note Beats.

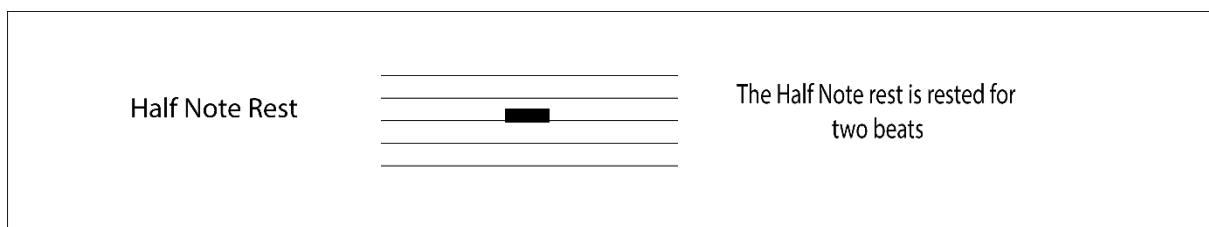


Figure 9. Half Note Rest Value



Alien Guitar Secrets

QUARTER NOTE RESTS

Quarter Note Rests are a block of time you DONOT play for, and which last for exactly One Quarter Note Beat.

Quarter Note Rest		The Quarter Note rest is rested for one beat
-------------------	---	--

Figure 10. Quarter Note Rest Value

EIGHTH NOTE RESTS

Eighth Note Rests are a block of time you DONOT play for, and which last for exactly Half of One Quarter Note Beat.

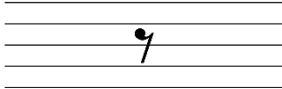
Eighth Note Rest		The Eighth Note rest is rested for 1/2 of one beat
------------------	---	--

Figure 11. Eighth Note Rest Value

SIXTEENTH NOTE RESTS

Sixteenth Note Rests are a block of time you DONOT play for, and which last for exactly One Quarter of One Quarter Note Beat.

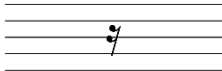
Sixteenth Note Rest		The Sixteenth Note rest is rested for 1/4 of one beat
---------------------	---	---

Figure 12. Sixteenth Note Rest Value



Alien Guitar Secrets

TRIPLETS

Another type of common notational is called a Triplet. Triplets give a piece of music the feeling of movement or momentum. By its name, the number three (3) is at its heart.

‘Still Got the Blues’ (6/8) by Gary Moore, and ‘Since I Been Loving You’ (12/8) by Led Zeppelin, are two examples of the triplet feel.

Eighth Note Triplets are indicated by three eighth notes beamed together and given the value of three eighth notes inside of one quarter note as follows:

Quarter Note
=
3 Eighth Notes



1 & a

Triplet Eighth Notes three eighth notes made to fit a quarter note and counted as follows:



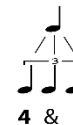
1 & a



2 & a



3 & a



4 & a

Figure 13. Triplet eighth Note Value

HOW TO COUNT BEATS

Counting beats is not that hard. It just takes a little practice done regularly. Every note value is worth a specific number of beats because the note themselves are a subdivision of a fixed marker. In this case it's the quarter note that is fixed and it is valued at one beat.

So, we say that there are four quarter note beats to a whole note, two quarter note beats to a half note, one beat to a quarter note, $\frac{1}{2}$ beat to an eighth note, $\frac{1}{4}$ beat to a sixteenth note etc.

To better understand this, we count each type of note in a unique way, like it has its own signature.



Alien Guitar Secrets

Counting Whole Notes

1 2 3 4

1 2 3 4

Counting Half Notes

1 2 3 4

1 2 3 4

Counting Quarter Notes

1 2 3 4

1 2 3 4

Counting Eighth Notes

1 & 2 & 3 & 4 &

1 & 2 & 3 & 4 &

Counting Sixteenth Notes

1 e & a 2 e & a 3 e & a 4 e & a

1 e & a 2 e & a 3 e & a 4 e & a

Counting Triplet Eighth Notes

1 & a 2 & a 3 & a 4 & a

1 & a 2 & a 3 & a 4 & a

Figure 14. Counting The Beat



Alien Guitar Secrets

TIME SIGNATURES

Now that we have a better understanding of what Note and Rest values are, and how to count them, we now need to put them into a context of structure, so we can create real musical artifacts with them.

For a piece of music to be organised, so both players and audience alike feel as if they understand what is being presented as coherent, music require a countable and measurable format to be able to rely upon.

We call the framework for this time structure a 'Time Signature'. Along with Key Signatures which tell you what key to play in, Time Signatures, instruct us as to how to count the rhythmic time structure the piece of music is to played in.

Time Signatures are super easy to understand once you get your head around the terms and annotations pertaining to the various directions given in a piece of music.

Music Notes are written in Bars or Measures, as they are also known. The value given to each bar is the Time Signature of a piece of music and each Measure is divided by what we call Bar Lines.

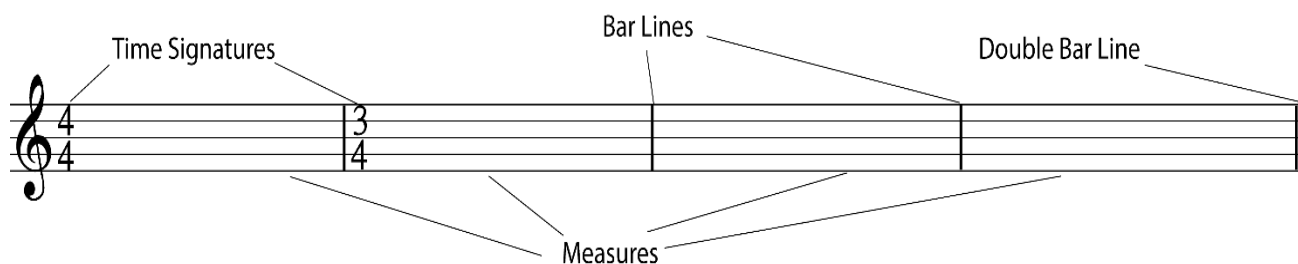


Figure 15. Time Indicators on a Music Staff



Alien Guitar Secrets

The most common of Time Signatures is **4/4** or unsurprisingly also known as 'Common Time'. This looks like a Mathematical fraction, and that's because, it kind of is. Understanding what the numbers mean, one atop the other is crucial in understanding how to count a piece of music.

So, what do the numbers mean? Again, looking at it like a mathematical fraction is the easiest and most correct way.

We can take the top number and look at it as simply as indicating the number of beats in the bar. But what value do each of the beats in the bar abide to?

This is what the bottom number means.

Now, when this concept of time signatures was presented to me as a kid, I struggled to understand what this bottom number meant. But all it's just the feel of the notes in the bar which is being determined by the value of this bottom number. So, if we have a time signature of say **4/4**, this means there are **4 beats to the bar**, and the **value given to each of these beats is a quarter note**. Now **3/4** would mean there are **3 beats** at **quarter note value** to the bar.

Now, if the number on the bottom is an **8**, it means that now **each beat in the bar is given the value of an eighth Note**. So, **6/8** would be **6 beats** to the bar at **eighth note value**, and **9/8** will **9 beats** to the bar at **eighth note value**.

Time Signatures

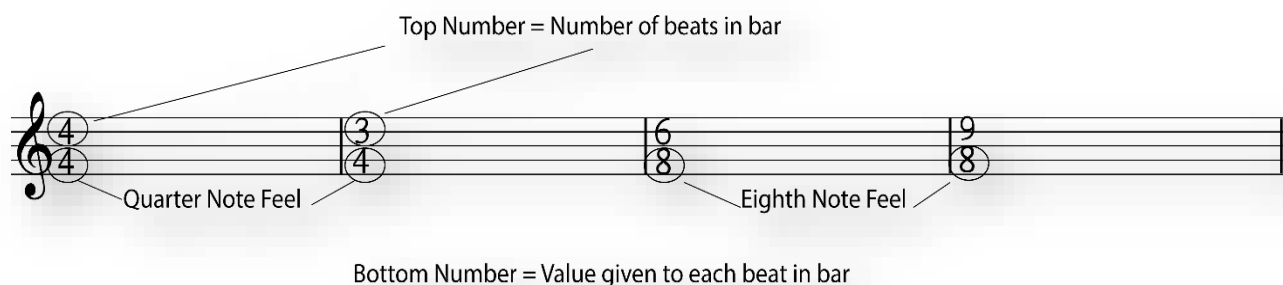


Figure 16. Time Signatures



Alien Guitar Secrets

TIME SIGNATURES IN ACTION

Time Signatures

The figure displays three musical staves, each with four measures of music and counting patterns below them.

- Staff 1 (4/4):**
 - Measure 1: Four quarter notes (1, 2, 3, 4).
 - Measure 2: Quarter, eighth, quarter, eighth (1, 2 & 3, 4 &).
 - Measure 3: Quarter, eighth, eighth, quarter, eighth, eighth (1 & 2 & a 3 & 4).
 - Measure 4: Quarter, eighth, eighth, quarter, eighth, eighth (1 2 e & 3 & 4).
- Staff 2 (3/4):**
 - Measure 1: Three quarter notes (1, 2, 3).
 - Measure 2: Quarter, eighth, quarter, eighth (1 & 2, 3 &).
 - Measure 3: Quarter, eighth, eighth, quarter, eighth, eighth (1 e & 2 & 3 &).
 - Measure 4: Quarter, eighth, eighth, quarter, eighth, eighth (1 & a 2, 3).
- Staff 3 (6/8):**
 - Measure 1: Six eighth notes (1 2 3 4 5 6).
 - Measure 2: Quarter, eighth, quarter, eighth (1 _ 3 4 5 6).
 - Measure 3: Three eighth notes, eighth, quarter, eighth, eighth, quarter (1 2 3 4 5 6).
 - Measure 4: Six eighth notes (1 2 3 4 5 6).

Figure 17. Common Time Signature Examples

As you can see above, if you understand the value of each of the notes or rests and be mindful of the number of beats in the bar, understanding time signatures is straight forward.

On the other hand, it can get really complex like this next example below from Dream Theater's 'Dance of Eternity'. I dare you to try and count this one.



Alien Guitar Secrets

DREAM THEATER - THE DANCE OF ETERNITY EXCERPT

dim. 136 137 138 139 140

141 142 143 144

♩ = 124

145 146 147 wide flam. 148 149

♩ = 177

150 151 152

153 154 155

156 157 158 159

Figure 18. Dream Theater - The Dance of Eternity Excerpt

ESSENTIAL RHYTHMIC PATTERNS

Understanding how rhythm works can be tricky at first, but it is vital we understand rhythm, so the practice of counting and memorising common rhythmic patterns is a must.



Alien Guitar Secrets

RHYTHMIC PATTERN PRACTICE

Rhythmic Pattern 1



Rhythmic Pattern 2



Rhythmic Pattern 3



Rhythmic Pattern 4



Rhythmic Pattern 5



Rhythmic Pattern 6



Figure 19. Essential Rhythmic Patterns



Alien Guitar Secrets

STRAIGHT V SWING FEEL

We have already discussed that note and rest values are the musical fraction system of dividing notes and rests into smaller pieces of time so we can adjust the speed of a piece of music into shapes and rhythmic patterns.

The most basic notation of time is what we call simple time. This is where a note is divisible by two. The feel for Simple Time is what we call a Straight Feel.

Triplets have more of a rolling feel to them, and are the division of one beat into three equal beats.

But they are still straight in their feel

A Swing feel is characterised by pushed or Swung feel. The difference between straight or swing beats lies in the feel.

Take the Peter Gunn Theme Intro below:

This is what I would deem to be a straight eighth note shuffle beat.



Alien Guitar Secrets

Further to the standard triplet feel is what we call Swung Triplets.

When playing straight eighth notes, we are dividing the beats exactly in half; where each note lasts an equal length of time.

In contrast with swung eighth notes, the second eighth note in each pair occurs two-thirds of the way through the beat. This is equivalent to playing on the first and third hits of an eighth note triplet.

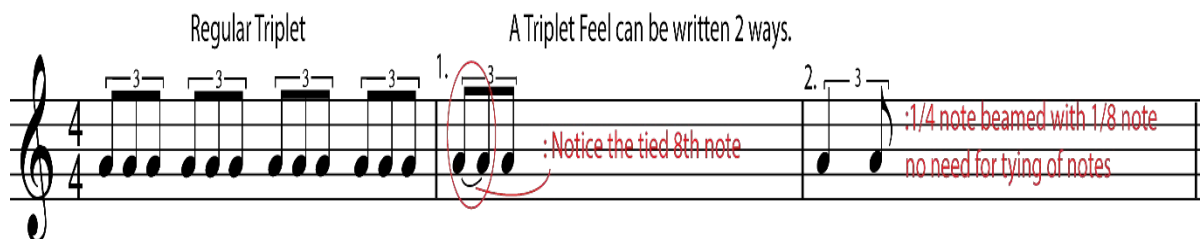


Figure 20. Regular Triplet and Swung Triplet Feel

If we hit on beat 1 of the triplet, hold for the duration of beat 2 (this is called a tied note), and then play again on beat 3, this then gives a feeling of swing because it creates a longer duration for the note on beat 1 and a shorter duration for the note on beat 3.



Alien Guitar Secrets